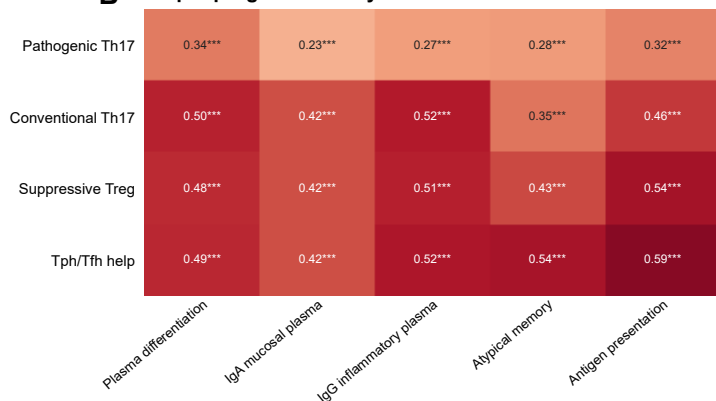


Participant-level helper/regulatory and cytotoxic programs covary with B-cell states in IBD

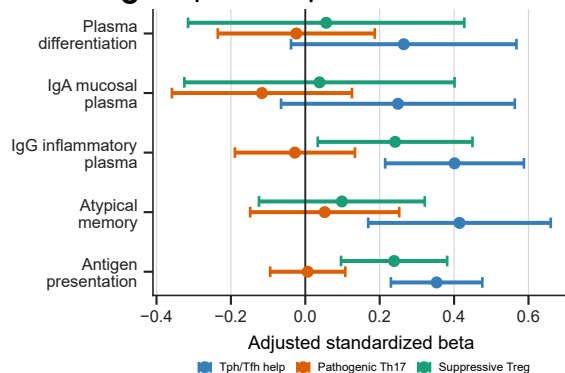
A Global repertoire and program covariation



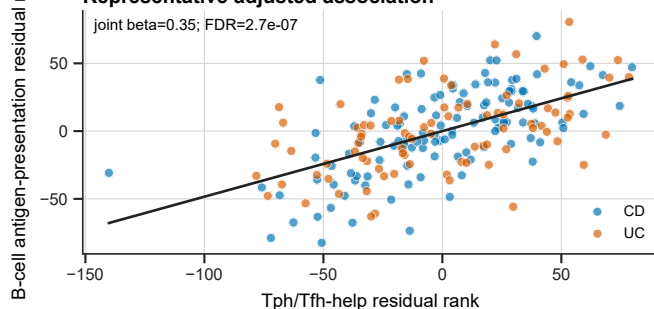
B Helper programs covary with B-cell states in IBD



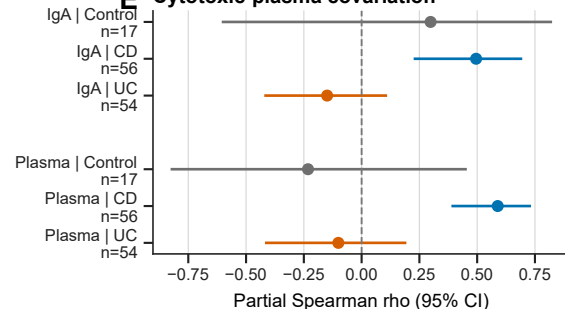
C Independent helper contributions



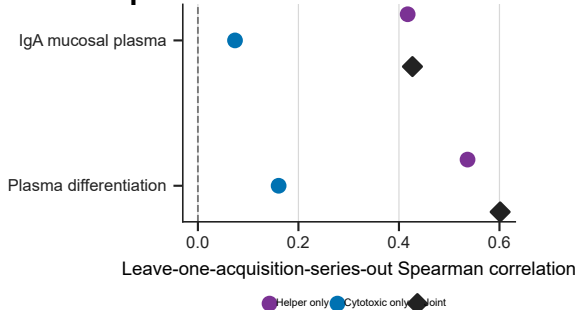
D Representative adjusted association



E Cytotoxic-plasma covariation



F Leave-one-series-out robustness



G Bidirectional helper-B-cell signaling

